

MASLYANINO, Russian Federation

WMO: 297360

Lat: 54.336N

Lon: 84.208E

Elev: 199

StdP: 98.96

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.0	(c) -35.4	(d) -41.8	(e) 0.1	(f) -38.8	(g) -38.4	(h) 0.1	(i) -35.3	(j) 10.4	(k) -4.9	(l) 9.4	(m) -6.0	(n) 0.1	(o) 0	(p) 0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.5	(c) 29.3	(d) 18.9	(e) 27.7	(f) 18.3	(g) 26.0	(h) 17.5	(i) 20.6	(j) 26.1	(k) 19.6	(l) 25.2	(m) 18.7	(n) 24.3	(o) 3.2	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.9	23.1	17.7	13.0	22.0	16.6	12.1	21.0	60.7	26.2	57.1	25.1	54.2	24.4	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.9	(b) 7.5	(c) 6.4	(e) -41.5	(f) 32.0	(g) 3.9	(h) 1.5	(i) -44.3	(j) 33.1	(k) -46.6	(l) 34.0	(m) -48.7	(n) 34.8	(o) -51.5	(p) 36.0		
			DB													
			WB													
				(e) -41.5	(f) 22.6	(g) 3.8	(h) 1.4	(i) -44.2	(j) 23.6	(k) -46.4	(l) 24.5	(m) -48.6	(n) 25.3	(o) -51.3	(p) 26.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.1	-18.8	-15.4	-6.7	2.8	10.5	16.3	18.4	15.6	9.3	2.5	-7.1	-15.2		
	DBStd	14.20	9.07	7.78	6.44	5.27	4.90	3.95	2.99	3.48	4.09	4.98	7.87	8.95		
	HDD10.0	3998	894	710	518	223	56	4	0	2	61	234	514	781		
	HDD18.3	6364	1152	943	777	467	245	81	37	98	271	490	764	1039		
	CDD10.0	753	0	0	0	6	71	195	261	176	41	3	0	0		
	CDD18.3	76	0	0	0	0	2	21	39	14	1	0	0	0		
	CDH23.3	1115	0	0	0	3	104	338	418	222	30	0	0	0		
	CDH26.7	257	0	0	0	0	22	76	97	54	7	0	0	0		
(14) Wind	WSAvg	2.3	2.1	2.3	2.4	2.6	2.6	2.1	1.8	1.7	1.9	2.4	2.6	2.5		
(15) Precipitation	PrecAvg	479	24	20	21	27	43	61	68	55	44	42	40	34		
	PrecMax	600	47	47	49	67	96	122	148	96	90	102	98	72		
	PrecMin	348	4	0	2	5	9	18	22	10	15	21	16	11		
	PrecStd	75	11	11	11	13	25	28	32	24	19	19	18	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	1.5	9.0	22.6	29.4	30.7	31.5	31.1	27.1	19.2	9.3	1.9		
		MCWB	0.2	0.2	3.8	12.4	16.2	18.7	19.8	19.6	15.9	11.4	6.7	0.4		
	2%	DB	-1.1	-0.4	6.0	18.7	26.1	28.7	29.2	28.1	22.8	15.8	6.5	0.2		
		MCWB	-1.8	-1.8	2.4	10.4	14.6	18.1	19.9	18.9	14.6	9.3	4.1	-0.8		
	5%	DB	-3.3	-2.1	3.9	15.6	23.8	27.1	27.4	25.8	20.2	13.0	3.9	-1.4		
		MCWB	-4.2	-3.2	1.2	8.7	13.5	17.8	19.2	17.6	13.1	8.1	2.3	-2.3		
	10%	DB	-5.6	-4.0	2.1	12.0	20.9	25.2	25.7	23.7	17.4	10.3	2.0	-3.1		
		MCWB	-6.5	-4.9	0.1	6.7	12.5	17.2	18.5	16.7	11.9	6.9	0.8	-4.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	0.6	4.4	13.1	17.4	21.0	22.3	21.2	17.1	11.8	7.0	0.7		
		MCDB	1.2	1.2	8.1	20.7	26.3	27.0	27.7	27.2	23.2	17.0	8.9	1.5		
	2%	WB	-1.9	-1.6	2.6	11.1	16.0	19.5	21.2	19.7	15.3	10.0	4.2	-0.7		
		MCDB	-1.1	-0.6	5.3	17.7	24.3	26.0	26.4	25.8	21.5	14.4	5.9	0.2		
	5%	WB	-4.2	-3.2	1.5	9.1	14.4	18.7	20.2	18.6	14.0	8.5	2.4	-2.3		
		MCDB	-3.4	-2.2	3.5	14.2	21.3	25.2	25.3	24.6	19.3	12.4	3.9	-1.5		
	10%	WB	-6.5	-4.9	0.2	7.1	13.1	17.7	19.2	17.5	12.5	7.0	0.8	-4.0		
		MCDB	-5.6	-3.9	2.1	11.9	19.9	23.5	24.2	22.6	16.3	10.2	1.8	-3.1		
(35) Mean Daily Temperature Range	MDBR		10.5	12.2	12.2	11.5	13.5	12.6	11.5	12.4	12.0	9.5	7.8	9.1		
	5% DB	MCDBR	8.4	10.5	12.9	18.2	19.2	17.0	15.0	16.5	18.1	15.6	7.7	8.2		
		MCWBR	8.2	9.7	9.2	10.4	9.0	7.3	6.5	7.7	9.8	9.5	5.8	7.8		
	5% WB	MCDBR	8.5	10.1	10.8	16.7	16.7	14.2	12.6	14.3	16.1	14.0	7.4	8.0		
		MCWBR	8.3	9.6	8.2	10.1	8.6	6.7	6.3	7.1	9.1	9.1	6.0	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.334	0.384	0.414	0.413	0.413	0.438	0.414	0.377	0.360	0.336	0.304		
	taud		1.995	1.939	1.873	2.035	2.177	2.263	2.194	2.246	2.293	2.207	2.014	1.940		
	Ebn at Noon		597	706	757	803	832	836	803	799	781	688	536	483		
	Edh at Noon		75	113	152	148	136	127	134	120	100	86	74	65		
(44) All-Sky Solar Radiation	RadAvg		1.00	2.04	3.46	4.62	5.37	6.09	5.49	4.62	3.17	1.60	0.81	0.65		
	RadStd		0.09	0.13	0.30	0.23	0.59	0.47	0.35	0.33	0.35	0.21	0.14	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page